# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

| | | | |
|---|---|---|---|
| **Course Code:** | CSA1715 | **Course Title:** | Artificial Intelligence |
| **CO Assessed:** | CO1 – Apply (BL3) | **Assessment:** | Assessment Tool 1 – Analytical Problem Solving |
| **Weightage:** | 35% of CIA | **Duration / Marks:** | 60 mins |
| **CO1 Statement:** | Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3) | | |
| **Student Name:** | S.YAMINI | **Reg. No.:** | 1973260015 |
| **Section / Batch:** | 2026-30 | **Date:** | 03/10/2026 |

### Code of Conduct

*I S.YAMINI, 1973260015 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: S.YAMINI

**Instructions**

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
- Duration: 1 hour (15 minutes per question).

| Section / Topic | Focus | Question Nos. |
|---|---|---|
| **1. Foundations and History of AI** | Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI | **Q3** |
| **2. Intelligent Agents: Agents and Environments** | Agent function and program, PEAS description, task environment properties, agent architectures | **Q1, Q2** |
| **3. Concept of Rationality** | Performance measure, percept sequence, expected utility, rational versus omniscient agent | **Q1, Q4** |

**Annexure A – Analytical Problem Solving**

## Question 1: Two-Room Vacuum World

*Topic: Agents and environments, concept of rationality | 15 minutes*

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

**Performance measure over a lifetime of T = 6 time steps:** +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take p = 1 unless told otherwise).

- **Agent X (simple reflex):** if the current square is Dirty, Suck; otherwise move to the other square.
- **Agent Y (reflex with internal state):** if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

**(a)** Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. **[4 Marks]**

Performance: +1 for every clean square at the end of each time step; −p for every Left/Right move. Lifetime T = 6 steps.

Environment: Two-room vacuum world with squares A and B; initially both are Dirty and dirt never reappears.

Actuators: Suck, Right, Left, NoOp.

Sensors: Current location and whether the current square is Dirty or Clean.

**(b)** Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for p = 1. **[5 Marks]**

Agent X rule: Dirty → Suck; Clean → move to the other square. Starting at A, both squares are Dirty.

Step 1: At A (Dirty) → Suck. Clean squares = 1. Move penalty = 0. Step score = 1.

Step 2: At A (Clean) → Right. Clean squares = 1. Move penalty = 1. Step score = 0.

Step 3: At B (Dirty) → Suck. Clean squares = 2. Move penalty = 0. Step score = 2.

Step 4: At B (Clean) → Left. Clean squares = 2. Move penalty = 1. Step score = 1.

Step 5: At A (Clean) → Right. Clean squares = 2. Move penalty = 1. Step score = 1.

Step 6: At B (Clean) → Left. Clean squares = 2. Move penalty = 1. Step score = 1.

Total score for p = 1 = 1 + 0 + 2 + 1 + 1 + 1 = 6

**(c)** Trace Agent Y in the same way and find its total score for p = 1. **[4 Marks]**

Agent Y uses internal state. It remembers when both squares have been cleaned; after that it performs NoOp.

Step 1: At A (Dirty) → Suck. Clean squares = 1. Move penalty = 0. Step score = 1.

Step 2: At A (Clean) → Right. Clean squares = 1. Move penalty = 1. Step score = 0.

Step 3: At B (Dirty) → Suck. Clean squares = 2. Move penalty = 0. Step score = 2.

Step 4: Both squares remembered clean → NoOp. Clean squares = 2. Move penalty = 0. Step score = 2.

Step 5: NoOp. Clean squares = 2. Move penalty = 0. Step score = 2.

Step 6: NoOp. Clean squares = 2. Move penalty = 0. Step score = 2.

Total score for p = 1 = 1 + 0 + 2 + 2 + 2 + 2 = 9.

**(d)** Derive the total score of X and of Y as functions of p. For which values of p is Y strictly better than X? **[3 Marks]**

Agent X makes four moves (Steps 2, 4, 5 and 6), so its total score is:

$X(p) = 7 - 4p$.

Agent Y makes only one move (Step 2), so its total score is:

$Y(p) = 10 - p$.

Y is strictly better when $10 - p > 7 - 4p$.

Therefore, $3p > -3$, so $p > -1$.

For the stated non-negative penalty domain $p \geq 0$, Y is strictly better for every $p \geq 0$.

**(e)** Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. **[4 Marks]**

Agent Y is rational under this performance measure because it uses its percept history/internal state to stop moving after both rooms are known to be clean. This avoids unnecessary movement penalties while preserving the maximum cleanliness reward.

Agent X cannot match Y because X is a simple reflex agent: it reacts only to the current percept. When it sees a Clean square, it must move to the other square, even though it has no useful reason to keep moving after both squares are clean. Y has memory of past percepts, so it can select NoOp. Thus, identical current percepts can lead to different rational actions when the percept history is different.

## Question 2: Task Environment Analysis and Agent Selection

*Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes*

Three AI agents are to be deployed in the following task environments:

- **E1:** An autonomous taxi driving through city traffic.
- **E2:** A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- **E3:** A chess-playing agent in a tournament game played with a clock.

**Difficulty score convention:** for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

| Property | Easier value (0) | Harder value (1) |
|---|---|---|
| **Observability** | Fully observable | Partially observable |
| **Number of agents** | Single agent | Multi-agent |
| **Determinism** | Deterministic | Stochastic |
| **Episodic / Sequential** | Episodic | Sequential |
| **Static / Dynamic** | Static | Dynamic (semi-dynamic = 0.5) |
| **Discrete / Continuous** | Discrete | Continuous |

**(a)** Classify each environment on all six properties with a one-phrase justification for each. **[9 Marks]**

E1 – Autonomous taxi:

• Observability: Partially observable — traffic, intentions and hazards are not fully visible.

• Number of agents: Multi-agent — other drivers, pedestrians and road users act independently.

• Determinism: Stochastic — traffic events and other agents introduce uncertainty.

• Episodic/Sequential: Sequential — current driving decisions affect future states.

• Static/Dynamic: Dynamic — the traffic situation changes while the agent operates.

• Discrete/Continuous: Continuous — position, speed and steering vary continuously.

E2 – Conveyor-belt part-picking robot:

• Observability: Partially observable — camera view may be limited or noisy.

• Number of agents: Single agent — the robot is the decision-making agent.

• Determinism: Stochastic — belt motion, object positions and camera uncertainty vary.

• Episodic/Sequential: Sequential — actions affect later picking opportunities.

• Static/Dynamic: Dynamic — parts continuously move on the conveyor.

• Discrete/Continuous: Continuous — object position and motion vary continuously.

E3 – Chess-playing agent:

• Observability: Fully observable — the board state is visible to both players.

• Number of agents: Multi-agent — two competing players make decisions.

• Determinism: Deterministic — legal moves produce predictable board transitions (ignoring clock/environmental issues).

• Episodic/Sequential: Sequential — every move changes future possibilities.

• Static/Dynamic: Semi-dynamic — the board does not change by itself, but the clock changes while thinking; score = 0.5.

• Discrete/Continuous: Discrete — board states and legal moves are discrete.

**(b)** Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. **[4 Marks]**

Using the given convention (harder value = 1; semi-dynamic = 0.5):

E1 Taxi = 1 + 1 + 1 + 1 + 1 + 1 = 6.0

E2 Robot = 1 + 0 + 1 + 1 + 1 + 1 = 5.0

E3 Chess = 0 + 1 + 0 + 1 + 0.5 + 0 = 2.5

Therefore, from the numerical difficulty score:

E1 = 6.0, E2 = 5.0, E3 = 2.5.

**(c)** For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. **[4 Marks]**

For the hardest environment E1 (autonomous taxi), a utility-based agent is suitable. The taxi must handle partial observability, multiple agents, stochastic events, sequential decisions, dynamic traffic and continuous variables. A utility-based architecture can compare alternatives using factors such as safety, travel time and route quality.

For the easiest environment E3 (chess), a goal-based agent is suitable. Chess has a fully observable, discrete, deterministic environment, and the agent can evaluate actions in relation to the goal of achieving checkmate/winning the game. In practice, search and evaluation are used to select moves.

**(d)** Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. **[3 Marks]**

The domain name alone does not determine program complexity. The environment properties determine what the agent must represent and reason about. Partial observability requires handling incomplete information; stochasticity requires uncertainty handling; sequential environments require planning; dynamic environments require continual updating; multi-agent environments require considering other agents; and continuous environments require processing continuously varying states. Therefore, E1 needs a more complex decision process than a simple fully observable, discrete environment.

## Question 3: Turing Test Data Across the History of AI

*Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes*

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

| System | Era / technique | Interrogations | Correctly identified |
|---|---|---|---|
| **ELIZA** | 1960s, keyword pattern matching | 50 | 47 |
| **Chatbot B** | 2010s, scripted rules plus statistical NLP | 50 | 36 |
| **System C** | 2020s, large neural language model | 50 | 29 |

**Reference:** Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

**(a)** Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. **[6 Marks]**

Correct-identification rate = (correctly identified as machine / 50) × 100.

Fooling rate = ((50 − correctly identified) / 50) × 100.

ELIZA: Correct = 47/50 × 100 = 94%; Fooling = 3/50 × 100 = 6%.

Chatbot B: Correct = 36/50 × 100 = 72%; Fooling = 14/50 × 100 = 28%.

System C: Correct = 29/50 × 100 = 58%; Fooling = 21/50 × 100 = 42%.

**(b)** Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. **[5 Marks]**

Method: compare each correct-identification rate with the 70% criterion and compare it with the 50% guessing level.

ELIZA: 94% > 70%, so it satisfies the criterion. Distance from guessing = 94 − 50 = 44 percentage points.

Chatbot B: 72% > 70%, so it satisfies the criterion. Distance from guessing = 72 − 50 = 22 percentage points.

System C: 58% < 70%, so it does not satisfy the criterion. Distance from guessing = 58 − 50 = 8 percentage points.

**(c)** Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. **[5 Marks]**

ELIZA fooling rate = 6%.

System C fooling rate = 42%.

Improvement from ELIZA to System C = 42 − 6 = 36 percentage points.

Ratio = 42/6 = 7. Thus, System C's fooling rate is 7 times ELIZA's.

Chatbot B fooling rate = 28%.

Improvement from Chatbot B to System C = 42 − 28 = 14 percentage points.

**(d)** Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument). **[4 Marks]**

The Turing Test evaluates whether a machine can act humanly in a text conversation; it does not directly measure internal reasoning or rationality. A high fooling rate means judges often classify the machine as human, but that result alone cannot establish that the system is thinking in the human sense.

Using the “acting humanly” approach, System C's 42% fooling rate is evidence about its conversational behavior in this test. It is not proof of human-like thought. The Chinese Room argument also illustrates the distinction: producing appropriate language responses does not, by itself, establish genuine understanding. Therefore, Turing-test performance should be interpreted as behavioral evidence within the specific test conditions.

## Question 4: Rational Route Choice under Uncertainty

*Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes*

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

$$U = 100 - T - 20 \cdot L$$

where T is the travel time in minutes and L = 1 if the taxi arrives after the 55-minute deadline, otherwise L = 0.

| Route | Normal time (min) | Probability of disruption | Time if disrupted (min) |
|---|---|---|---|
| **R1: Highway** | 30 | 0.30 (accident) | 70 |
| **R2: City road** | 45 | 0.10 (signal failure) | 60 |
| **R3: Ring road** | 50 | 0 (never disrupted) | 50 |

**(a)** Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. **[4 Marks]**

Formula: U = 100 − T − 20L, where L = 1 for arrival after 55 minutes and L = 0 otherwise.

R1 Highway:

• Normal: T = 30, on time → L = 0 → U = 100 − 30 = 70.

• Disrupted: T = 70, late → L = 1 → U = 100 − 70 − 20 = 10.

R2 City road:

• Normal: T = 45, on time → L = 0 → U = 100 − 45 = 55.

• Disrupted: T = 60, late → L = 1 → U = 100 − 60 − 20 = 20.

R3 Ring road:

• Normal: T = 50, on time → L = 0 → U = 100 − 50 = 50.

• Disrupted: no disruption → T = 50, on time → U = 50.

**(b)** Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose. **[5 Marks]**

Expected utility = P(normal)×U(normal) + P(disrupted)×U(disrupted).

R1: P(disruption)=0.30, so P(normal)=0.70.

EU(R1) = 0.70(70) + 0.30(10) = 49 + 3 = 52.

R2: P(disruption)=0.10, so P(normal)=0.90.

EU(R2) = 0.90(55) + 0.10(20) = 49.5 + 2 = 51.5.

R3: P(disruption)=0.

EU(R3) = 1.00(50) = 50.

Thus the expected utilities are R1 = 52, R2 = 51.5, R3 = 50. Under the stated utility measure and probabilities, the rational choice is R1 because it has the highest expected utility.

**(c)** A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. **[4 Marks]**

The highway disruption probability becomes 0.80, so P(normal) = 0.20.

New EU(R1) = 0.20(70) + 0.80(10)

= 14 + 8

= 22.

The other expected utilities remain R2 = 51.5 and R3 = 50. Therefore, with the new percept, R1 is no longer the highest-expected-utility route; R2 has the highest expected utility among the three.

**(d)** Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. **[4 Marks]**

Part (b) selected R1. If the disruption occurs, the taxi takes 70 minutes and arrives after the 55-minute deadline.

Actual utility = 100 − 70 − 20(1) = 10.

The choice in part (b) was not irrational merely because the unfavorable outcome occurred. Rationality is evaluated using the information available to the agent when it made the decision and the specified performance measure. An omniscient agent would know in advance that the accident would occur and could choose accordingly, but a rational agent does not have such perfect future knowledge. It chooses the action with the best expected performance based on its available percept sequence.

**(e)** Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. **[3 Marks]**

Rationality depends on both the percept sequence and the performance measure. The percept sequence determines what information the agent has when selecting an action, while the performance measure defines what counts as a good outcome.

In part (c), the roads and their normal/disrupted travel times did not change. The new radio alert changed the agent's percept by increasing the estimated probability of a highway disruption from 0.30 to 0.80. Consequently, the expected utility of R1 fell from 52 to 22, changing the rational decision. This demonstrates that rational action is information-dependent rather than based only on fixed route properties.

**Rubrics for Calculation**

| Criteria | Weightage | Level 4 – Exemplary (100% of marks) | Level 3 – Proficient (75%) | Level 2 – Developing (50%) | Level 1 – Beginning (25%) |
|---|---|---|---|---|---|
| **Problem Understanding** | **20%** | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| **Method Selection** | **20%** | Selects most appropriate method/formula with strong justification | Selects correct method with minor errors | Inappropriate or partially correct method | Unable to select method |
| **Solution Process** | **25%** | Logical, systematic steps with correct approach throughout | Mostly correct steps with minor mistakes | Disorganized steps; multiple errors | No clear process |
| **Accuracy of Calculation** | **20%** | All calculations accurate; correct final answer | Minor calculation errors | Frequent errors; incorrect final answer | No valid calculations |
| **Interpretation of Results** | **15%** | Clearly interprets results with units and engineering relevance | Basic interpretation with minor omissions | Limited or unclear interpretation | No interpretation |

| Q. No. / Criteria | Problem Understanding (4) | Method Selection (4) | Solution Process (5) | Accuracy of Calculation (4) | Interpretation of Results (3) | Sub. Total (20) | Total % |
|---|---|---|---|---|---|---|---|
| 1 | | | | | | | |
| 2 | | | | | | | |
| 3 | | | | | | | |
| 4 | | | | | | | |
| Total | | | | | | | |

| Faculty In-charge | Course Coordinator | Program Director |
|---|---|---|
| | | |
| Signature: _____ | Signature: _____ | Signature: _____ |
| Date: _____ | Date: _____ | Date: _____ |